

---

# **Database Project Report**

for

## **Attendance and Leave Management Portal**

**Version 1.0 approved by Adnan Aslam**

**Prepared by Ahmad 221775, Faraz 221847, Nimra Marrium 222679**

**BS Software Engineering, 4<sup>th</sup> Semester**

**Department of Creative Technology**

**28 May 2024**

## Table of Contents

---

|                                          |   |
|------------------------------------------|---|
| .....                                    | 1 |
| 1. Introduction .....                    | 3 |
| 2. Problem Statement.....                | 3 |
| 3. Project Selection Rationale .....     | 3 |
| 4. Tools and Technologies Used .....     | 4 |
| Backend.....                             | 4 |
| Frontend .....                           | 4 |
| Why these tools? .....                   | 5 |
| 5. Marketing Strategy .....              | 5 |
| 1. Promotion Strategy .....              | 5 |
| Social Media:.....                       | 5 |
| Email Marketing: .....                   | 6 |
| Word of Mouth:.....                      | 6 |
| Demonstrations and Tutorials: .....      | 6 |
| 2. Distribution Strategy: .....          | 6 |
| Educational Institutions: .....          | 6 |
| Online Communities:.....                 | 6 |
| 6. Targeted Audience .....               | 7 |
| Primary Audience: .....                  | 7 |
| Importance and Meeting Their Needs:..... | 7 |
| Teachers: .....                          | 7 |
| School Administrators: .....             | 7 |
| IT Staff: .....                          | 7 |
| 7. Effectiveness of the Project.....     | 8 |
| Anticipated Effectiveness:.....          | 8 |
| Measuring Success: .....                 | 8 |

## 1. Introduction

The Attendance and Student Leave Management Portal is an exhaustive portal system. The system's role is to store the students' attendance data and help teachers when they apply for leave. We propose to develop a portal for the management of attendance and students' absences to increase data storage efficiency and accessibility. To manage and record the attendance of students effectively, this portal should provide a solution for teachers

## 2. Problem Statement

In Higher educational institutions, managing the attendance of students and reviewing Students applications are the two tasks that take most of the time. Teachers and staff associated with portals are affected by the outdated methods and technology used, which result in errors, time-consuming load times, poor user interfaces, and a lack of accessibility to users. Modern techniques and modern software should be used to modernize the portal to make it more user-friendly

## 3. Project Selection Rationale

I chose to develop Student Attendance and Leave Management portal because portals are affected by the outdated methods and technology used, which result in errors, and

time-consuming load times. By using modern techniques and modern software we should make it more user-friendly and reliable

This project is of great relevance to my studies, since I am a fourth semester student in software engineering. This project directly uses the software design, user interface design, and database programming skills I've gained from my coursework. This project not only help me in education sector but also play an important part in growth of technical and problem-solving skill

## 4. Tools and Technologies Used

The following tools are used in the project's backed and server management:

1. Node.js
2. React.js
3. MySQL
4. XAMPP

### Backend

Used Node.js with usage of frameworks like express, axios and Cors. Axios is used for API'S, Cors is used as Middleware for the communication and express as framework for backend.

### Frontend

use React.js with SCSS, Vite and React Router Dom. Vite used to host development live stuff, to do at the live time and React Router Dom for the navigation of our website

## Why these tools?

Node.js is an ideal choice to develop any system as it offers several advantages like efficiency, non-blocking I/O, scalability, fast performance, single language for back-end, rich ecosystem, strong community support and asynchronous programming capabilities. The choice of Node.js to develop the backend of the system is based on these advantages

React.js is a powerful, robust and adaptable library for modern web development. Its basic publishing syntax, object-oriented architecture, and strong community support make it an excellent choice for this project

## 5. Marketing Strategy

The following methods shall be followed for the marketing of this project:

### 1. Promotion Strategy

#### Social Media:

- We will use free social media platforms like LinkedIn etc.
- By posting regular updates, feature highlights.

- We will share our project in educational groups and forums to reach teachers and school administrators.

### Email Marketing:

- By sending personalized emails to professors, department heads, and school administrators at local schools and universities.

### Word of Mouth:

- We will tell our friends, classmates, and professors about the project to spread the word
- Also, by offering to present the portal at faculty meetings or education-related events at our university

### Demonstrations and Tutorials:

- We will create simple tutorial videos and upload them to YouTube to explain how to use the portal

## 2. Distribution Strategy:

### Educational Institutions:

- We will reach out to local schools and universities to offer the portal for demo use
- We will provide training sessions or materials to help schools use the portal effectively

### Online Communities:

- We will share the portal on educational technology forums, social media groups, and other online communities where educators gather

## 6. Targeted Audience

### Primary Audience:

1. Teachers
2. School Administrators
3. IT Staff in Educational Institutions

### Importance and Meeting Their Needs:

#### Teachers:

*Importance:* Directly handle attendance and leave management.

Needs Met: Efficient attendance recording, user-friendly interface, real-time access to data.

#### School Administrators:

*Importance:* Oversee school operations and ensure accurate record-keeping.

Needs Met: Reliable attendance data, comprehensive reports, improved communication.

#### IT Staff:

*Importance:* Implement and maintain school software systems.

Needs Met: Easy implementation and maintenance, minimal technical support required, scalable solution.

## 7. Effectiveness of the Project

### Anticipated Effectiveness:

- **Impact:** The Attendance and Student Leave Management Portal is expected to significantly reduce the time and effort required for managing attendance and leave applications, leading to more efficient administrative processes
- **Adoption:** High potential for widespread adoption in educational institutions due to its user-friendly interface and comprehensive features
- **User Satisfaction:** By addressing the specific needs of teachers, administrators, and IT staff, the portal aims to improve overall satisfaction and ease of use
- **Data Accuracy:** Enhanced accuracy in attendance records and leave management, reducing errors

### Measuring Success:

- **User Feedback:** Collect feedback from teachers and administrators about their experience using the portal. Conduct simple surveys to gather opinions on ease of use, functionality, and overall satisfaction
- **Usage Metrics:** Track the number of users actively using the portal. Monitor the frequency of logins and the number of attendance records and leave applications processed
- **Adoption Rates:** Measure how many educational institutions and individual users have adopted the portal
- **Error Reduction:** Compare the number of errors in attendance and leave management before and after implementing the portal
- **Support Requests:** Track the number of support requests and issues reported by users to identify areas for improvement